Propagation Velocities

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Lecture Objective

By the end of this lecture, you will be able to:

- Understand the concept of group velocity of a modulated signal.
- Evaluate phase and group velocities.
- Interpret dispersion diagrams.

Phase Velocity

- The phase velocity of a time-harmonic (sinusoidal time dependence*) propagating wave is given by

$$u_p = \frac{\omega}{\beta}$$

* a.k.a. “monochromatic”

Information Carrying Signal (1 of 3)

- To send a message, we need to encode information into some property of the propagating wave (amplitude, frequency, or phase). This process is called **modulation**.
- Doing so, the wave is no longer monochromatic but occupies some bandwidth.
- Even simply turning the signal on and off – OOK modulation – results in the signal occupying some bandwidth.

Information Carrying Signal (2 of 3)

- A simple example of a modulated signal results from the multiplication of a low frequency gaussian pulse with a high frequency sinusoid.
- Mathematically, this modulated signal can be represented as a superposition of sinusoids, i.e., by a Fourier transform.

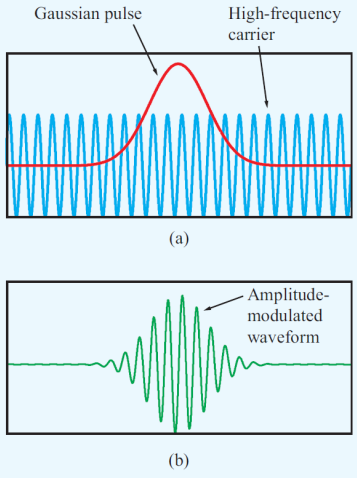
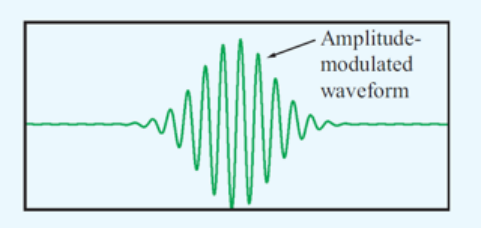
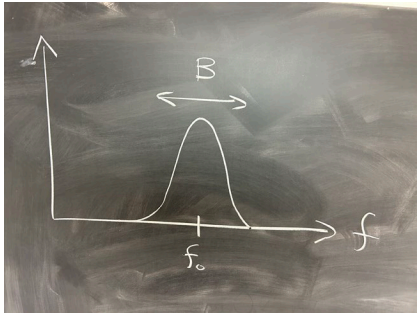


Figure 8-25 The amplitude-modulated high-frequency waveform in (b) is the product of the Gaussian-shaped pulse with the sinusoidal high-frequency carrier in (a).

Information Carrying Signal (3 of 3)



Signal in the time domain.



Signal in the frequency domain.

The modulated signal is referred to as a "wave group".

Question

The bandwidth of an information bearing signal is proportional to the baud (symbols per second) and also depends on the shape of the pulses used to represent the symbols. Let's assume that the bandwidth is given by $B = 1.2\ R_b$ where R_b is the baud and B is the resulting bandwidth.

Assume our signal has a baud of 2 MS/s (mega-symbols per second) and modulates a 2.45 GHz carrier wave.

What is the absolute bandwidth (in Hz) for this signal?

What is the fractional bandwidth (relative to the carrier frequency) for this signal?

Group Velocity

- The velocity with which the wave group propagates is called the "**group velocity**".
- For narrowband signals (i.e., $B << f_0$), the group velocity is given by

$$\boxed{u_g = \left.\frac{1}{d\beta / d\omega}\right|_{f=f_0}}$$

- For narrowband signals, the velocity with which information propagates is equal to the group velocity

Phase and Group Velocities of a Plane Wave in Simple Lossless Non-Magnetic Medium

- Phase constant:

$$\beta = \frac{\omega}{c}\sqrt{\varepsilon_r}$$

- Phase velocity:

$$u_p = \frac{\omega}{\beta} = \frac{c}{\sqrt{\varepsilon_r}}$$

- Group velocity:

$$u_g = \left.\frac{1}{d\beta/d\omega}\right|_{\omega=\omega_0} = \frac{1}{\sqrt{\varepsilon_r}/c} = \frac{c}{\sqrt{\varepsilon_r}}$$

Phase and Group Velocities of TE10 Wave in Air-Filled Rectangular Hollow-Tube Waveguide (1 of 2)

- Phase constant:

$$\beta = \frac{\omega}{c}\sqrt{1-\left(\omega_{10}/\omega\right)^2}$$

- Phase velocity:

$$u_p = \frac{c}{\sqrt{1-\left(\omega_{10}/\omega\right)^2}}$$

- Group velocity:

$$u_g = \left.\frac{1}{d\beta/d\omega}\right|_{\omega=\omega_0} = c\sqrt{1-\left(\omega_{10}/\omega\right)^2}$$

Phase and Group Velocities of TE10 Wave in Air-Filled Rectangular Hollow-Tube Waveguide (2 of 2)

- Notice that

$$u_p u_g = c^2$$

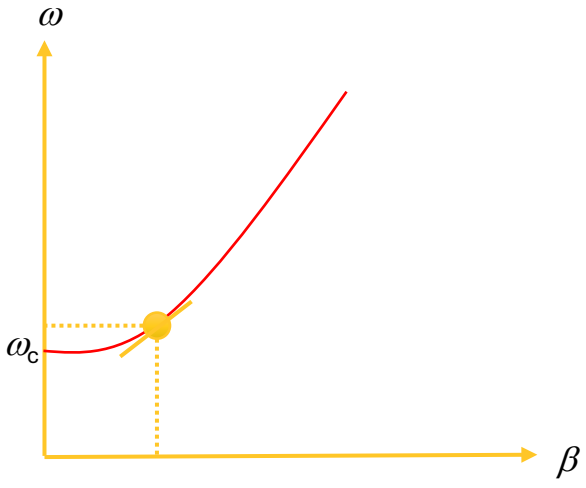
- Furthermore

$$u_p \xrightarrow{\omega \to \infty} c$$

$$u_g \xrightarrow{\omega \to \infty} c$$

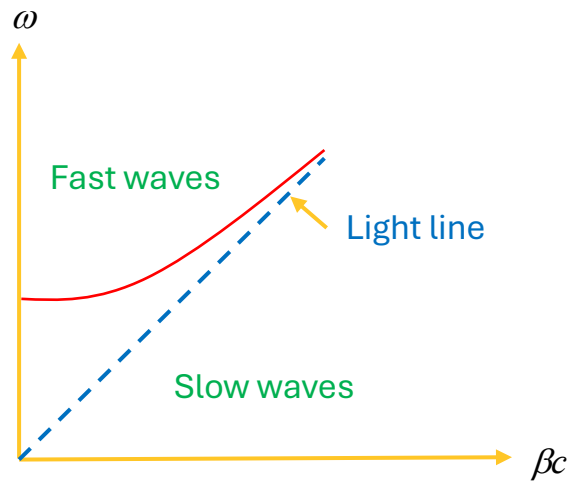
Dispersion (ω-β) Diagrams (1 of 2)

- A dispersion or ω-β diagram is a plot of frequency (ω) as a function of phase constant (β).
- The ratio of ω to β at a point on the curve is the phase velocity.
- The slope of the curve is the group velocity.
- The y-intercept is the cutoff frequency of the mode.



Dispersion (ω-β) Diagrams (2 of 2)

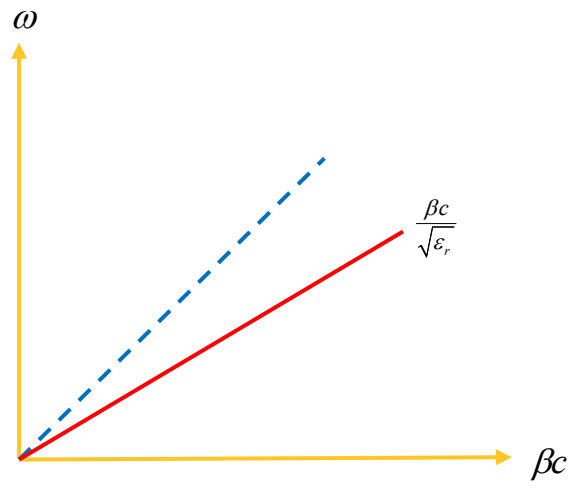
- Often, we plot ω versus β multiplied by c.
- Doing so, means that a line of slope 1 corresponds to points on which the phase velocity is equal to the speed of light in a vacuum. This line is often called the "light line".
- Points below the light line correspond to "slow waves" since $u_p < c$.
- Points above the light line correspond to "fast waves" since $u_p > c$.



Dispersion (ω-β) Diagram for TEM Mode in Lossless Simple Non-Magnetic Medium

$$\beta = \omega\sqrt{\mu\varepsilon}$$

$$\beta = \frac{\omega}{c}\sqrt{\varepsilon_r}$$

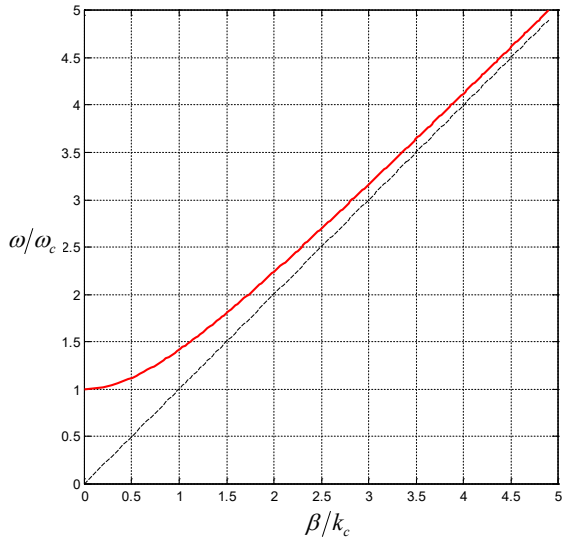


Dispersion (ω-β) Diagram for Mode in Hollow-Tube Waveguide

$$\beta = \frac{\omega}{c}\sqrt{1-(\omega_c/\omega)^2} = \frac{\omega_c}{c}\sqrt{(\omega/\omega_c)^2-1}$$

$$= k_c\sqrt{(\omega/\omega_c)^2-1}$$

$$\boxed{\frac{\beta}{k_c} = \sqrt{\left(\frac{\omega}{\omega_c}\right)^2-1}}$$



Zig-Zag Angle Interpretation of Fields in Waveguide (1 of 3)

- It is possible to interpret the fields of a given mode in a waveguide as those of plane waves bouncing back and forth in the guide.
- Consider the electric field of the TE10 mode in RWG:

$$\tilde{E}_y = E_0 \sin\left(\frac{\pi}{a}x\right) e^{-j\beta_{10} z}$$

Zig-Zag Angle Interpretation of Fields in Waveguide (2 of 3)

- We can re-write the above expression and represent the electric field as a superposition of two plane waves:

$$\tilde{E}_y = -j\frac{E_0}{2}\left(e^{j\frac{\pi}{a}x}e^{-j\beta_{10}z} - e^{-j\frac{\pi}{a}x}e^{-j\beta_{10}z}\right) = -j\frac{E_0}{2}\left(e^{j\kappa\sin\theta x}e^{-j\kappa\cos\theta z} - e^{-j\kappa\sin\theta}e^{-j\kappa\cos\theta z}\right)$$

- Where

$$\kappa\sin\theta = \frac{\pi}{a}$$

$$\kappa\cos\theta = \beta_{10} = \sqrt{k_0^{\,2} - \left(\frac{\pi}{a}\right)^2}$$

$$\kappa = k_0$$

$$\theta = \operatorname{atan}\left(\frac{\pi}{\beta_{10}a}\right)$$

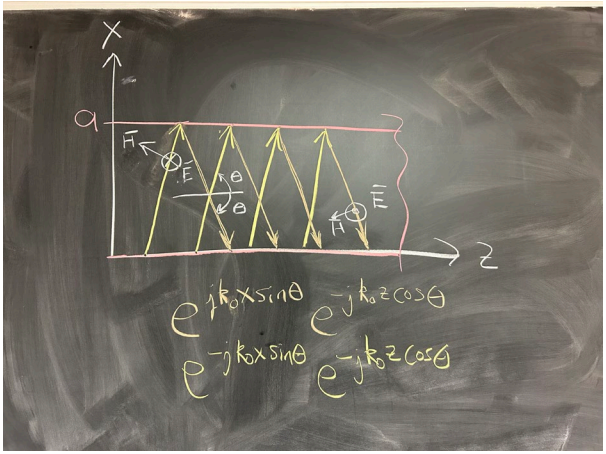
Zig-Zag Angle Interpretation of Fields in Waveguide (3 of 3)

- Hence:

$$\tilde{E}_y = -j\frac{E_0}{2}\left(e^{jk_0 x\sin\theta}e^{-jk_0 z\cos\theta} - e^{-jk_0 x\sin\theta}e^{-jk_0 z\cos\theta}\right)$$

- A superposition of two plane waves propagating at the speed of light at angles θ wrt the z-axis.
- Note that

$$\theta = \operatorname{atan}\left(\frac{\pi}{\beta_{10}a}\right) \xrightarrow{\omega\to\infty} 0$$



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